

OPENER: A Frugal Open-World NER Pipeline for the Legal Domain

Anonymous ACL submission

Abstract

Legal text is hard for named entity recognition. New statutes, parties and courts appear continually, and expert annotation is costly. Open-world NER assigns types not fixed at training time, but these systems are built on general-domain text and adapted per corpus and language for legal use. We built OPENER for this setting instead, a frugal pipeline of ready-to-use components whose entity embedder is deliberately left generic, so its space is never fitted to one legal schema. We evaluate its zero-shot head (OPENER-ZS) and supervised linear probe (OPENER-Sup) on four legal corpora in English, Portuguese and German, against GLiNER, GNER, OWNER and a 4-bit Qwen, reporting quality (Adjusted Mutual Information), latency and energy jointly over three seeds. The design holds under an English domain shift and degrades cross-lingually. OPENER-ZS stays usable in English (up to 49.1 gold AMI) but does not lead, weakening cross-lingually (23.8–37.4), whereas OPENER-Sup is barely affected (61.4 on average, on a par with the 62.3 of a preliminary out-of-domain check) and posts the best end-to-end score (35.9). Every system is then capped by the English detectors’ recall (as low as 10.2% on German), a ceiling both heads reach at one to two orders of magnitude less latency and energy than the generative baselines. With labels, the supervised head is the practical operating point, and detection, not typing, is where adaptation should go first. That it holds up this well motivates the general-domain study we are now running.

1 Introduction

Named Entity Recognition (NER) extracts entities such as persons, organisations and dates from raw text, and is a foundational step in most information extraction pipelines. Standard NER systems are closed-set: trained once on a fixed inventory of types, they can only predict those types. Real-

world applications rarely afford such a static inventory, and legal text is an extreme case. Entity inventories are open and shifting (new statutes, case citations, parties, courts, contractual instruments, jurisdictions), documents are long and formulaic, annotation requires expert effort, and extraction errors carry material consequences. Re-annotating and retraining a tagger for every new type or sub-domain is rarely practical.

This is precisely the regime that *open-world* NER targets: recognising and typing entities whose categories were not fixed at training time. Notable open-world systems take different routes to it: span scoring against type names (Zaratiana et al., 2024), generative tagging (Ding et al., 2024), a dedicated entity encoder clustered into dynamically named types (Genest et al., 2025), and prompted instruction-tuned models (Zhou et al., 2024), at far higher inference compute. All are trained and evaluated on general-domain text, and their legal uses adapt them per corpus, schema and language, by fine-tuning a tagger or by prompting a large model anew. Few legal deployments can pay that price for every schema, sub-domain and jurisdiction, so we take the opposite route: one open-world pipeline whose components are never fitted to a legal corpus, measured component by component to locate the bottleneck and show where the next annotation or compute budget should go.

We do this with OPENER, a frugal open-world pipeline we designed for the legal setting from ready-to-use pretrained components: a frozen open-world detector followed by a light contrastively fine-tuned, Matryoshka-truncatable entity embedder. Its embedder is fine-tuned on general-domain entity spans and never on a legal schema, a design choice rather than a shortcut: an open-world space must stay unspecialised, or it collapses into the closed-set taggers we are trying to avoid. It also runs one to two orders of magnitude cheaper in latency and energy than trained-encoder and gener-

ative baselines, which legal practice, rarely GPU-rich, needs. We evaluate its two operating points: OPENER-ZS, a label-free head (label-name prototypes, transductive refinement, detector fusion), and OPENER-Sup, a balanced linear probe fit on the legal target labels. Our contributions are:

- **A frugal pipeline for legal NER.** We assemble an open-world pipeline from pretrained components with *no* legal-domain training of any of them, and we test both of its typing heads, the label-free OPENER-ZS and the supervised OPENER-Sup.
- **A legal open-world benchmark.** Four legal NER corpora in three languages: English (SEC/EDGAR filings and Indian court judgements), Portuguese and German (Brazilian and federal court decisions). Eight systems are compared end to end under one protocol: both OPENER heads, the standard open-world baselines (GLiNER-S/M/L, GNER-T5 and OWNER), and a 4-bit Qwen2.5-1.5B.
- **A three-axis, seed-averaged protocol.** We report quality (AMI), latency and energy jointly rather than accuracy alone, and average every OPENER score over three embedder-retraining seeds.
- **An explanation of why the two heads diverge.** OPENER-ZS stays usable on English legal text (up to 49.1 gold AMI) but does not lead the zero-shot field, trailing GLiNER-M/L and, on average, OWNER. OPENER-Sup is barely dented by the domain shift (61.4 mean gold AMI, on a par with the 62.3 the same head reaches in a preliminary general-domain check) and posts the best end-to-end score (35.9), because its balanced probe draws boundaries between closely related legal types that a single label-name match cannot. Both heads are then capped by the English detectors’ recall (down to 18.7% on German for GLiNER-L, 10.2% for GLiNER-M), which makes in-language *detection*, not typing, the first place further adaptation should go. Keeping the embedding space unspecialised therefore costs little, which is what motivates the general-domain study of the same pipeline we are now carrying out.

2 Related Work

2.1 Open-world and zero-shot NER

The three open-world systems we compare against differ in backbone as much as in approach: GLiNER (Zaratiana et al., 2024) scores spans against candidate type names with a DeBERTaV3 encoder (He et al., 2023), GNER (Ding et al., 2024) generates them with a sequence-to-sequence model (Raffel et al., 2020), and OWNER (Genest et al., 2025) trains a dedicated entity encoder, then clusters mentions into dynamically named types. Typing from type descriptions (Aly et al., 2021) needs no labelled example at all. A complementary line distils or prompts billion-parameter language models for open NER, as in UniversalNER (Zhou et al., 2024), GoLLIE (Sainz et al., 2024) and ChatIE (Wei et al., 2023), at a compute and energy cost our comparison also probes via a small 4-bit Qwen2.5-1.5B baseline. OPENER differs from all of these by reusing a pretrained embedding rather than training an encoder, which is what keeps it cheap enough for the legal setting we target.

2.2 Legal NLP and legal NER

Legal NLP is an active field spanning summarisation, judgement prediction, question answering and information extraction, as a recent legal-NLP survey documents (Ariai et al., 2025), driven by the specialised vocabulary, long formulaic documents and jurisdiction-specific entity inventories of legal text. On the modelling side, domain-adapted language models such as LEGAL-BERT (Chalkidis et al., 2020) and multi-task benchmarks such as LexGLUE (Chalkidis et al., 2022) show that in-domain pretraining and evaluation matter for legal NLU.

The dominant approach to legal NER is to train or adapt a supervised tagger for each corpus, schema and language. LegNER (Karamitsos et al., 2025) domain-adapts BERT with span supervision for English legal NER and anonymisation. A Serbian legal NER study (Kalušev and Brkljač, 2025) fine-tunes a BERT-type model for court rulings and releases a new dataset. The LegalEval shared task (SemEval-2023 Task 6) spurred systems such as a Bi-LSTM with Flair embeddings for Indian court judgements (Ramesh and Eswara, 2023). Even LLM-based studies remain schema-bound: a financial NER study (Lu and Huo, 2025) prompts GPT-4o, LLaMA-3.1 and Gemini-1.5, and reports systematic failure modes on domain enti-

ties. These systems achieve strong in-corpus scores but each requires labelled data and (re)training tied to one schema and one language. The underlying corpora expose exactly such schema-specific types: E-NER (Au et al., 2022) annotates U.S. SEC/EDGAR filings with seven classes including legal-specific ones (court, government body, legislation/act), LeNER-Br (Luz de Araujo et al., 2018) covers Brazilian court decisions with dedicated *legislation* and *case-law* tags, and German-LER (Leitner et al., 2020) annotates German federal decisions with a fine-grained 19-class schema.

Crucially, the E-NER study (Au et al., 2022) reports that NER models trained on general English degrade sharply on legal text, precisely the distribution shift we study. Rather than retraining a legal-specific tagger per corpus and language, we build a single open-world pipeline that no legal schema ever touches, and measure how far it goes on all four of them, positioned against the standard open-world systems and, where applicable, domain baselines (Section 4).

2.3 Energy-aware evaluation

Reporting efficiency alongside accuracy is increasingly advocated, from the Green AI agenda (Schwartz et al., 2020) to energy audits of NLP training (Strubell et al., 2019) and systematic carbon reporting (Henderson et al., 2020). The scope runs from the carbon cost of training (Patterson et al., 2022) to a whole-lifecycle view of AI’s environmental footprint (Wu et al., 2022), and to inference-time cost specifically, where a deployment-cost study (Luccioni et al., 2024) shows general-purpose generative models to be orders of magnitude more expensive per query than task-specific ones, precisely the gap this frugal pipeline targets. We adopt the same three-axis (quality/latency/energy) view, measured with the CodeCarbon package (Courty et al., 2024) and reported in the energy-times-grid-intensity accounting of Lacoste et al. (2019).

3 Method

OPENER is built for the legal domain, yet none of its components is fitted to it. The detector and the entity embedder are ready-to-use checkpoints kept frozen, so no gradient step is ever taken on legal text. Only the typing head touches the legal side, and it does so differently at each operating point. OPENER-ZS consumes no legal label at

all, deriving its prototypes from the type names and refining them on the unlabelled test mentions. OPENER-Sup fits a linear classifier on the frozen embeddings of the target’s training mentions, so it consumes legal labels without updating a single encoder weight. We therefore measure how far a frugal open-world pipeline goes on legal NER *without* being retrained on it, which is the regime a legal deployment actually faces.

3.1 Mention detection and embedding

Spans are proposed by a frozen GLiNER detector (Zaratiana et al., 2024) (GLiNER-L for OPENER-ZS, GLiNER-M for OPENER-Sup, see Section 4.2) and embedded *in context* by Nomic v1.5 (Nussbaum et al., 2025), a Sentence-BERT-style (Reimers and Gurevych, 2019), Matryoshka-truncatable embedder (Kusupati et al., 2022) that we sharpen once, with a FaceNet triplet step (Schroff et al., 2015) on English, general-domain CoNLL-2003 (Tjong Kim Sang and De Meulder, 2003) spans and an error-driven hard-negative mining pass. Training it away from any legal schema is deliberate, and the same single checkpoint then serves all four legal schemas here without re-fitting. Nothing in the pipeline is legal-specific except the type inventory it is asked about, and that is exactly the design this paper puts to the test.

Formally, the detector maps a sentence s to a set of candidate mentions $\mathcal{M}(s) = \{m_i\}$, each mention is embedded *in context*, and a typing head assigns it a type drawn from the target inventory $\mathcal{T}$,

$$e_i = f_\theta(s, m_i) \in \mathbb{R}^D, \quad \hat{y}_i \in \mathcal{T}, \quad (1)$$

where f_θ is the frozen embedder and e_i the D -dimensional vector it produces for mention m_i , read together with its sentence s . We keep the full $D=768$. The predicted type $\hat{y}_i$ is drawn from $\mathcal{T}$, the legal type inventory of the target corpus, which f_θ never saw during its own training.

3.2 Entity typing

OPENER-ZS The zero-shot head, the open-world default, uses no target labels. Each legal type t is represented by a prototype p_t , the ℓ_2 -normalised centroid of the embeddings of the words forming its name,

$$\bar{e}_t = \frac{1}{|\text{name}(t)|} \sum_{w \in \text{name}(t)} f_\theta(w), \quad (2)$$

$$p_t = \bar{e}_t / \|\bar{e}_t\|_2,$$

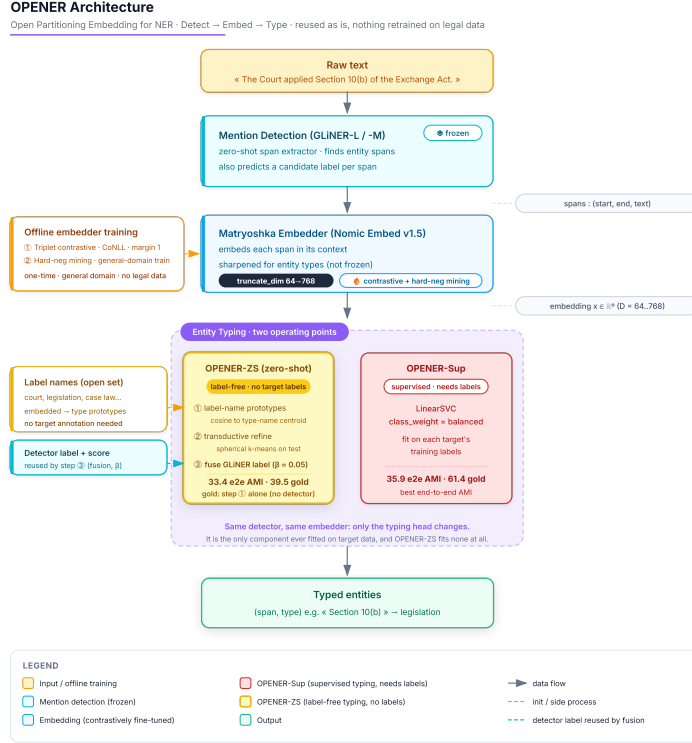


Figure 1: The OPENER pipeline (Detect → Embed → Type), with no component fitted to legal data, with its two typing heads: OPENER-ZS (label-name prototypes, transductive refinement, detector fusion) and OPENER-Sup (supervised linear probe). Nothing is retrained on legal data. The legal AMI figures shown are means over the three embedder-retraining seeds. The end-to-end ones use the full OPENER-ZS head, the typing-on-gold ones its first step alone, since no detector is in the loop (Section 4.2).

where $\text{name}(t)$ is the set of anchor words naming type t , for instance “court” or “case law citation”. Each anchor word w contributes its embedding $f_\theta(w)$, and $\bar{e}_t$ is their average. Dividing that average by its Euclidean norm $\|\bar{e}_t\|_2$ gives a unit-length prototype. Equation (2) is the typing-on-gold form. End-to-end, each anchor is first embedded in four short templates whose vectors are averaged in place of the bare $f_\theta(w)$, which changes nothing below. These anchor words are English for every corpus, which is precisely what the cross-lingual results of Section 4.3 put under stress.

A mention is assigned the nearest prototype by cosine similarity, refined by a *transductive* pass of spherical k -means over the unlabelled test mentions, seeded by the prototypes, and *fused* with the detector’s own zero-shot label,

$$s_t(i) = \cos(e_i, \tilde{p}_t) + \beta g_i \mathbb{1}[t = \ell_{\text{GLINER}}(i)], \quad (3)$$

where $s_t(i)$ is the fused score of assigning type t to mention m_i , and $\tilde{p}_t$ the centroid of type t after that refinement, which is p_t itself when the transductive pass is switched off. The detector contributes the type $\ell_{\text{GLINER}}(i)$ it predicted for that span and its confidence $g_i \in [0, 1]$ in it. The indicator $\mathbb{1}[\cdot]$

is worth 1 only when t is that predicted type, so the fusion term only ever raises the score of the detector’s own choice, and the mention is typed as $\hat{y}_i = \arg \max_t s_t(i)$. The weight $\beta=0.05$ is fixed a priori and never tuned on legal data, so the detector only nudges the arg-max (Section 4.4 sweeps it post-hoc).

OPENER-Sup A balanced linear classifier (scikit-learn’s LinearSVC (Pedregosa et al., 2011) with `class_weight=balanced`, built on LIBLINEAR (Fan et al., 2008)) is fit one-vs-rest on the legal target’s training labels. For each type c , it minimises a squared-hinge objective with balanced class weights,

$$\min_{w_c, b_c} \frac{1}{2} \|w_c\|_2^2 + C \sum_i \omega_{y_i} \max(0, 1 - z_{ic} h_c(e_i))^2, \quad (4)$$

with $h_c(e_i) = w_c^\top e_i + b_c$ the one-vs-rest hyperplane of type c , e_i the frozen embedding of training mention i , y_i its gold legal type, $z_{ic} = +1$ if $y_i = c$ and -1 otherwise, and C the regularisation strength. The balancing weight $\omega_c = n/(|\mathcal{T}| n_c)$ up-weights

rare types, with n the total number of training mentions and n_c the count of type c . This matters on legal schemas, where a single type can hold well under 1% of the mentions (Court in E-NER, Table 1). A mention is typed as $\hat{y}_i = \arg \max_c h_c(e_i)$. Rather than a single label-name match, this carves the embedding space into one region per type, separating types whose vectors sit close together as long as their regions do not overlap. This is the supervised upper bound, and it needs target labels.

3.3 Evaluation

We report both *typing-on-gold* (perfect detection) and *end-to-end* protocols. Predicted and gold mentions are aligned by exact character offsets, and detection errors stay in the evaluation through sentinel labels, one for a gold mention no system predicted and one for a predicted span with no gold counterpart, following the alignment protocol of Genest et al. (2025). Sentences carrying no gold entity are not evaluated, on any system. Quality is the Adjusted Mutual Information (AMI) (Vinh et al., 2010) between the gold type partition U and the predicted partition V over the aligned mentions,

$$\text{AMI} = \frac{\text{MI} - \mathbb{E}[\text{MI}]}{\frac{1}{2}[H(U) + H(V)] - \mathbb{E}[\text{MI}]}, \quad (5)$$

where $\text{MI} = \text{MI}(U, V)$ is the mutual information between the two partitions, which measures how much a mention’s predicted type reduces the uncertainty about its gold type, and H the entropy. Subtracting the chance agreement $\mathbb{E}[\text{MI}]$ makes a random typing score near 0 whatever the number of types, a perfect one 1. The denominator uses the arithmetic-mean normalisation (the scikit-learn default).

We deliberately keep AMI as the single quality axis. A type-level F1 presupposes a fixed correspondence between predicted and gold type names, which is exactly what an open-world system does not provide: OWNER names its clusters dynamically, the generative baselines paraphrase or invent type strings, and a prototype head answers with the type whose *name* sits closest, not with a member of the gold label set. AMI compares the two partitions of the mentions instead, so predicted and gold names never have to coincide, and it is corrected for chance, which matters when the four schemas range from 6 to 19 types of very unequal frequency. We also report per-sentence latency (p50) and energy (Wh per 1000 sentences, measured with CodeCarbon (Courty et al., 2024)).

4 Experiments

4.1 Datasets

We evaluate on four legal-specific NER sets across three languages that, together, separate two kinds of distribution shift. *Two are in English*, isolating a *pure domain shift* from the general-domain spans the embedder was fine-tuned on: E-NER (Au et al., 2022) (annotated U.S. SEC/EDGAR filings, 7 types) and Indian-Legal (Kalamkar et al., 2022) (Indian court judgements, 14 fine-grained legal types). LeNER-Br (Luz de Araujo et al., 2018) (Brazilian court decisions, Portuguese) and German-LER (Leitner et al., 2020) (German federal court decisions) add a *cross-lingual* shift. All four annotate *legal-specific* entity types (courts, statutes, legislation, case law, provisions), rather than the generic person/organisation/date schema of general NER. Table 1 summarises them with the entity-type distribution of their test mentions.

4.2 Setup

Embedder, detector and heads are exactly those of Section 3, and nothing is retrained on legal data: Nomic v1.5 at $D=768$ in span-in-context mode, reused as released, behind a frozen GLiNER detector at threshold 0.3 (GLiNER-L for OPENER-ZS, GLiNER-M for OPENER-Sup, whose probe is fit on its own train detections). The baselines are GLiNER-S/M/L, GNER-T5-base, OWNER and a 4-bit Qwen2.5-1.5B-Instruct, all prompted zero-shot with the target type names. Test splits are capped at the first 1000 entity-bearing sentences, with Qwen on a 200-sentence subsample.

Table 5 in Appendix A gives the complete configuration, including the typing hyperparameters, all fixed a priori and never tuned on a legal set: an English anchor dictionary of one to five words per type, embedded in four short templates before averaging, three transductive k -means iterations, $\beta=0.05$, and $C=1$ for the balanced linear probe. Evaluation code, anchor dictionaries and per-run logs are released anonymously with the submission.

The two protocols do not exercise the same zero-shot head. End-to-end, OPENER-ZS is the full head of Section 3, templated prototypes, transductive refinement and detector fusion. Typing on gold there is no detector in the loop, hence no label to fuse, so the head reduces to the plain nearest-prototype classifier: Table 2 measures the label-name geometry on its own, Table 3 the operating

Table 1: Legal NER datasets.

Dataset	Lang	Sent.	Ment.	Entity types (share % of test mentions)
E-NER (Au et al., 2022)	en	877	1,793	Business 41.8, Misc 18.3, Legislation/Act 13.5, Location 11.3, Person 7.6, Government 7.1, Court 0.3
Indian-Legal (Kalamkar et al., 2022)	en	866	3,028	Lawyer 16.7, Court 9.3, Respondent 8.8, Other-person 8.7, Provision 8.5, Statute 7.3, Date 7.1, Petitioner 6.0, then GPE, Precedent, Org, Judge, Case-number and Witness, from 5.9 down to 1.9
LeNER-Br (Luz de Araujo et al., 2018)	pt	636	1,536	Organização 32.6, Legislação 24.6, Pessoa 15.2, Tempo 12.5, Jurisprudência 12.0, Local 3.1
German-LER (Leitner et al., 2020)	de	1,000 on 3,018	1,740	Legal-norm 38.3, Case-law 23.4, Organization 15.0, Regulation 6.7, Person 6.1, Legal-literature 5.9, Location 4.7

Test sentences carrying ≥ 1 entity, and the mentions actually evaluated. “1,000 on 3,018” marks the only split above our 1000-sentence cap, every score for it being computed on those first 1000. Sentences with no gold entity are evaluated by no system. Each type is followed by its share (%) of test mentions. German-LER is shown with its 7-type *coarse* schema (19-type *fine* schema in Section 4.4).

point one would deploy.

We report two orthogonal sources of uncertainty. The embedder’s *training* is stochastic, so the whole study is repeated under three retraining seeds, every OPENER cell giving mean $\pm$ std, the baselines being frozen deterministic checkpoints run once. Inference is deterministic for a fixed embedder, so the remaining uncertainty is the finite test set, quantified with a paired bootstrap in Section 4.3. Efficiency is deliberately *not* seed-averaged: a seed changes the embedder’s weights, not its per-sentence FLOPs, so any spread would reflect the machine’s thermal state.

4.3 Results

Keeping the two protocols apart, Table 2 reports typing-on-gold AMI and Table 3 the end-to-end comparison on all three axes. OWNER appears in both, as the one baseline that types mentions it is given rather than detecting and typing jointly. Three findings stand out.

Label-free typing is strong in English but degrades cross-lingually. OPENER-ZS is usable on the two English sets (47.7 on E-NER and 49.1 on Indian, from English label-name prototypes alone), but it drops to 37.4 (pt) and 23.8 (de). Its English prototypes align well with English legal mentions, far less with other-language ones (Table 2). It also trails OWNER’s own label-free typing on all four sets (44.7 mean against 39.5), which trains an entity encoder rather than reusing one, so the label-free operating point is where reuse costs the most. The zero-shot head is where the cross-lingual distribution shift bites hardest. It is also the *seed-sensitive* head: its gold scores vary by ± 3 –6 AMI across the three embedder retrainings (the released one is in fact a low draw on the English sets),

whereas the supervised probe varies by at most ± 1.4 , because label-name prototypes depend on the fine geometry of the embedding space, which shifts from one contrastive retraining to the next, while a fitted probe absorbs those shifts.

Supervised typing transfers well, including across languages, because the separation it learns is geometric, not just statistical. Fitting a balanced linear probe on the same frozen, general-domain embedder reaches 61.4 mean typing-on-gold AMI over the four sets (70.9 and 60.8 on the two English corpora, and still 53.7 and 60.3 on Portuguese LeNER-Br and German-LER), and is stable across the embedder-retraining seeds (std ≤ 1.4 AMI). It leads OWNER on every set, by 16 AMI on E-NER and by 35 on German. It scores 62.3 in a preliminary general-domain check (ongoing work), so keeping the space unspecialised costs under a point of AMI here: a balanced probe fits hyperplanes precise enough to pull apart legal types whose vectors sit *close together* (a lawyer from a judge, a statute from a case citation), which a single label-name match cannot do without supervision. The probe adapts to whatever the embedder encodes, so target supervision largely absorbs the cross-lingual gap at typing time.

End-to-end, detection sets the ceiling, and the label-free head cannot pull ahead of the detector it fuses with. On these legal-specific schemas every system clusters in a narrow ≈ 33 –36 AMI band (Table 3), because the English detectors recover only a fraction of gold mentions (GLiNER-L offset recall 18.7–57.4%, GLiNER-M 10.2–61.1%). With so few mentions detected, no typing head can pull far ahead, and OPENER-ZS’s fusion essentially reverts to the GLiNER-L labels.

Table 2: Typing-on-gold AMI ($\times 100$, $\uparrow$).

System	E-NER (en)	Indian (en)	LeNER-Br (pt)	German (de)	Mean
OWNER [§]	54.5	59.0	39.7	25.7	44.7
OPENER-ZS	47.7 ± 3.5	49.1 ± 3.0	37.4 ± 6.0	23.8 ± 4.3	39.5 ± 3.0
OPENER-Sup*	70.9± 1.2	60.8± 1.4	53.7± 1.2	60.3± 1.0	61.4± 0.4

Typing on gold mentions, detection bypassed. OPENER cells: mean $\pm$ std over three embedder-retraining seeds (Section 4.2), test-set bootstrap std ± 0.8 –1.7 AMI. Best per column in bold.

* Only OPENER-Sup uses target labels. [§] OWNER typing its own clusters, single deterministic run of the released checkpoint, zero-shot transfer as in Table 3.

Per-dataset winners are specific (OWNER (Genest et al., 2025) on E-NER, GNER-T5 on German), though OWNER degrades most cross-lingually at several times the inference cost of either OPENER head, and Qwen trails throughout while being the most energy-hungry system per sentence. *Averaged over the four sets, OPENER-Sup posts the best end-to-end AMI (35.9), narrowly ahead of GLiNER-L (35.0) and GLiNER-M (34.9), for every retraining seed. OPENER-ZS (33.4) sits just below OWNER (33.7), well under the 39.5 the same fused head reaches in our preliminary general-domain runs, where it leads GLiNER-L’s 38.9 (out-of-domain end-to-end figures, unrelated to the legal gold mean of Table 2). The margins are small, as expected when detection is the ceiling, yet a paired bootstrap on the released model ($K=1000$) makes OPENER-Sup’s lead statistically significant (+1.0 AMI over both GLiNER-M and GLiNER-L, 95% CI [+0.4, +1.7] and [+0.1, +1.9]), while OPENER-ZS’s fusion sits significantly below GLiNER-L (−1.3, [−1.8, −0.8]), largely deferring to the detector’s labels. The typing head is therefore not the weak link, consistent with the typing-on-gold results.*

Table 4 consolidates the three axes into one row per system, with the model size each one carries. Both heads match the frugal GLiNER family in cost, one to two orders of magnitude below the generative baselines and several times below OWNER, and OPENER-Sup does it while leading on AMI, at 337M parameters against OWNER’s 294M and Qwen’s 1.5B. Quality and cost are therefore not traded off against each other here, which keeps OPENER deployable on modest hardware whichever head is chosen.

4.4 Analysis

Two distribution shifts act together here, and the results separate them cleanly. Under a pure *do-main* shift (E-NER), the deliberately unspecialised embedder already places legal types far enough

Table 3: Open-world legal NER, end-to-end: quality, latency and energy.

Model	E-NER (en)	Indian (en)	LeNER-Br (pt)	German (de)	Avg
<i>AMI ($\times 100$, $\uparrow$), zero-shot</i>					
GLiNER-S	33.4	33.9	33.7	31.1	33.0
GLiNER-M	33.9	34.8	34.8	36.1	34.9
GLiNER-L	34.7	35.2	36.3	34.0	35.0
GNER-T5	28.7	28.6	33.2	39.2	32.4
Qwen-1.5B [‡]	24.9	28.1	24.9	14.5	23.1
OWNER [§]	38.4	33.3	34.8	28.4	33.7
OPENER-ZS	33.4 ± 0.2	33.8 ± 0.5	33.5 ± 1.2	32.8 ± 0.4	33.4 ± 0.4
<i>AMI, with target labels</i>					
OPENER-Sup*	35.7 ± 0.3	35.2 ± 0.3	34.1 ± 0.7	38.5 ± 0.3	35.9± 0.4
<i>Latency, median per sentence (ms, $\downarrow$)</i>					
GLiNER-S	20	22	24	22	22
GLiNER-M	36	35	37	39	37
GLiNER-L	65	88	92	78	81
GNER-T5	2897	15784	8768	5705	8289
Qwen-1.5B [‡]	4852	8068	5929	2857	5427
OWNER [§]	328	755	420	398	475
OPENER-ZS	104	140	172	115	133
OPENER-Sup*	78	102	85	73	84
<i>Energy per 1000 evaluated sentences (Wh, $\downarrow$)</i>					
GLiNER-S	0.4	0.5	0.5	0.5	0.5
GLiNER-M	0.7	0.8	0.8	0.7	0.7
GLiNER-L	1.3	1.7	1.8	1.6	1.6
GNER-T5	49.2	161.0	106.3	67.3	96.0
Qwen-1.5B [‡]	80.5	157.7	(2886) [¶]	63.3	100.5
OWNER [§]	10.0	23.1	12.9	12.2	14.6
OPENER-ZS	2.1	3.3	3.4	2.1	2.7
OPENER-Sup*	1.4	2.4	1.6	1.3	1.7

Detector mentions, offset/sentinel alignment (Section 4.2). Best per column in bold, within each block, ties included, the two AMI sub-blocks bolded apart since OPENER-Sup is the only row that consumes target labels. OPENER AMI rows are mean $\pm$ std over three retraining seeds, baselines are frozen checkpoints with a single deterministic run. Latency and energy are measured on the released seed-42 run. For CO₂, count 0.056 gCO₂ per Wh, the 56 gCO₂/kWh grid intensity CodeCarbon applied to every run here. * Only OPENER-Sup uses target labels, every other row is zero-shot. [‡] Qwen AMI on a 200-sentence subsample. [§] OWNER cost is amortised wall-clock, not a per-call latency. Model sizes in Table 4.

[¶] A value in parentheses is left out of its row’s Avg. Here an isolated CodeCarbon over-count (Qwen/LeNER-Br), so that Avg uses the other three sets.

Table 4: Quality vs efficiency, four-set means of Table 3.

Model	Params	AMI $\uparrow$	p50 (ms) $\downarrow$	Energy (Wh) $\downarrow$
<i>Zero-shot (no target labels)</i>				
GLiNER-S	50M	33.0	22	0.5
GLiNER-M	200M	34.9	37	0.7
GLiNER-L	330M	35.0	81	1.6
GNER-T5	220M	32.4	8289	96.0
Qwen-1.5B [‡]	1.5B	23.1	5427	100.5
OWNER [§]	≈ 294 M	33.7	475	14.6
OPENER-ZS	467M	33.4 ± 0.4	133	2.7
<i>With target labels</i>				
OPENER-Sup*	337M	35.9± 0.4	84	1.7

End-to-end, energy per 1000 evaluated sentences (multiply by 0.056 for gCO₂). Best per column in bold, within each sub-block. Params count the whole pipeline, detector plus the 137M embedder, so OPENER-Sup is lighter than OPENER-ZS only because it runs on GLiNER-M rather than GLiNER-L.

* Only OPENER-Sup uses target labels. [‡] Qwen AMI on a 200-sentence subsample, its energy over three sets (Table 3). [§] OWNER cost is amortised wall-clock.

apart to be separated, by a label-name prototype (47.7 gold AMI, no target label at all) or, far more precisely, by a supervised hyperplane (70.9). A cross-lingual shift widens the gap, the label-free head dropping to 24–37 while the probe stays high (53.7/60.3) by re-fitting on target labels.

A specialised-schema stress test isolates fineness of separation from language. On German-LER’s fine 19-type schema OPENER-ZS falls further to $22.7_{\pm 2.1}$ gold AMI, as fine German type codes rarely have a faithful English label-name prototype, while OPENER-Sup still reaches $61.5_{\pm 0.7}$, *on a par with its coarse score* (60.3) despite nearly three times as many classes. The embedder separates even fine legal distinctions once a probe names them, so the zero-shot weakness is not the embedding geometry but the missing supervised signal.

The binding constraint end-to-end is detection, not typing: GLiNER-L’s offset recall falls from 57.4% (en) to 18.7% (de), and GLiNER-M’s, which OPENER-Sup runs on, from 61.1% to 10.2%, so the scores compress to a ≈ 33 –36 band whatever the typing head, within which OPENER-Sup still posts the best average (35.9) by re-typing the detected mentions with supervision.

Language alone does not explain that fall: GLiNER-L recovers 43.9% of the Portuguese mentions, on a par with the 44.5% it reaches on English Indian judgements, and only German collapses. Its schema is as singular as its language, since legal-norm and case-law citations make up 62% of its mentions and are long multi-token strings, while our alignment credits a mention only on an exact match of both character offsets. Cross-lingual transfer and long-span detection are therefore confounded there, and separating them takes a boundary-relaxed alignment we leave to future work. The reading is the same either way: an in-language *detector* would lift the ceiling far more than a better typing head. Within our frugal, retraining-free budget, OPENER-Sup is therefore the practical default wherever even a modest labelled sample exists, and OPENER-ZS a no-label fallback for target languages close to the embedder’s own.

Fusion weight β . The weight β balances the label-name centroids against the detector’s own labels. It is fixed a priori at 0.05, and sweeping it post-hoc on the released model, a pure inference knob that leaves the embedder untouched, tells the same story: at $\beta=0$ the refined prototypes alone reach 32.4 average AMI (30.5 without the transductive pass), and as β grows the head interpolates up to the GLiNER-L labels (35.0), a plateau reached by $\beta \approx 0.2$. The label-free head can therefore do no better end-to-end than the detector it fuses with. The sweep also exposes a component that does not survive the transfer: at $\beta=0.05$, dropping the trans-

ductive pass would *raise* the average from 33.7 to 34.8 (released seed, hence the offset from the 33.4 of Table 3). Re-estimating prototypes on test mentions helps when those are mostly true entities, as they are in our general-domain runs, whereas here two thirds of the detected spans are alignment noise and the centroids drift towards it. We keep the released configuration rather than tune it on legal test sets, but under a domain and language shift the label-free components are the ones to re-examine first.

5 Conclusion

We built OPENER for legal NER, a frugal open-world pipeline assembled from ready-to-use pre-trained parts and deliberately kept free of any legal-specific training, and evaluated both of its typing heads on four legal corpora in three languages against the standard open-world baselines.

The two heads diverge sharply on legal text. OPENER-ZS remains a usable label-free option on English legal text, but it does not lead the zero-shot field the way it does in our preliminary general-domain runs: here it trails GLiNER-M/L and OWNER on average, on gold mentions as well as end-to-end, and its gold AMI collapses cross-lingually as its English-centric prototypes stop matching target-language mentions. OPENER-Sup is almost unaffected, its gold AMI within a point of that out-of-domain check even on German-LER’s fine 19-type schema, and it posts the best end-to-end AMI on average, a statistically significant lead. The reason is geometric: a balanced probe fits hyperplanes precise enough to pull apart legal types whose vectors sit close together, which a single label-name match cannot do without supervision. The supervised head is therefore the safer operating point wherever labels exist, and OPENER-ZS is best reserved for languages close to the embedder’s own.

End-to-end quality remains bounded first by mention *detection*, whichever typing head runs behind it. Adapting the detector in-language, rather than the embedder, is the natural next step and preserves the frugal decoupled design.

Nothing in the design is tied to legal text beyond the type names it is asked about, and its supervised head loses under a point of AMI on schemas its embedding space never saw. Legal NER was the demanding case we built it for, and it holds, so the same three-axis study on general-domain open-

world NER is now under way.

Limitations

Test splits are capped at their first 1000 entity-bearing sentences, which affects German-LER only, and sentences carrying no gold entity are excluded, so precision on entity-free text is never measured. OPENER scores are means over three embedder-retraining seeds: the supervised and end-to-end scores are stable ($\text{std} \leq 1.4$ AMI), but the label-free prototypes vary by up to ± 6 AMI across retrainings, so single-seed zero-shot numbers should be read with care. Inference is deterministic for a fixed embedder, and test-set bootstraps on the released model confirm both the typing-on-gold scores ($\text{std} \pm 0.8$ – 1.7 AMI) and the significance of OPENER-Sup’s end-to-end lead, stable across resampling seeds 42/7/123. More fundamentally, both the embedder and the detector are English-centric and used without any legal- or language-domain adaptation, which is the design under test here and also its main exposure. The detectors’ low cross-lingual recall (on German, 18.7% for GLiNER-L and 10.2% for GLiNER-M) caps end-to-end quality and is the first thing a follow-up should address.

Ethics Statement

This work runs and evaluates entity recognition on four publicly released legal NER benchmarks (E-NER over SEC/EDGAR filings, an Indian court-judgement set, LeNER-Br, and German-LER), used only for evaluation and under their original licenses. Legal documents contain personal data such as the names of parties, judges and counsel. We introduce no new annotations and release no data, we only run inference over the existing public test splits, and we report no individual-level content, so the study adds no privacy exposure and makes no attempt at re-identification.

Open-world NER is error-prone, and our own results show recall dropping sharply outside English (as low as 10.2% on German). The pipeline is therefore meant as an assistive tool that surfaces candidate entities for human review, not as a component of automated legal decision-making, where a missed or mistyped entity can carry material consequences. A missed entity is silent, so a human in the loop remains necessary, especially cross-lingually. Finally, a central motivation of this study is energy frugality. We report per-sentence latency

and energy alongside accuracy precisely to make the environmental cost of deployment explicit, and to argue for models that are cheap enough to run on modest hardware.

The authors used Claude Code (Anthropic) to scaffold and debug the evaluation scripts and to draft and copy-edit this manuscript. The research question, the experimental design, the choice of baselines and datasets, the execution of all experiments and the interpretation of the results are the authors’ own. The authors reviewed and edited all content and take full responsibility for it.

References

- Rami Aly, Andreas Vlachos, and Ryan McDonald. 2021. Leveraging type descriptions for zero-shot named entity recognition and classification. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (ACL-IJCNLP)*, pages 1516–1528.
- Farid Ariai, Joel Mackenzie, and Gianluca Demartini. 2025. Natural language processing for the legal domain: A survey of tasks, datasets, models and challenges. *ACM Computing Surveys*. To appear; arXiv:2410.21306.
- Ting Wai Terence Au, Vasileios Lamos, and Ingemar Cox. 2022. E-NER — an annotated named entity recognition corpus of legal text. In *Proceedings of the Natural Legal Language Processing Workshop 2022 (NLLP)*, pages 246–255. Association for Computational Linguistics.
- Ilias Chalkidis, Manos Fergadiotis, Prodromos Malakasiotis, Nikolaos Aletras, and Ion Androutsopoulos. 2020. LEGAL-BERT: The muppets straight out of law school. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 2898–2904. Association for Computational Linguistics.
- Ilias Chalkidis, Abhik Jana, Dirk Hartung, Michael Bommarito, Ion Androutsopoulos, Daniel Martin Katz, and Nikolaos Aletras. 2022. LexGLUE: A benchmark dataset for legal language understanding in English. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4310–4330. Association for Computational Linguistics.
- Benoit Courty, Victor Schmidt, Sasha Luccioni, et al. 2024. CodeCarbon: Estimate and track carbon emissions from machine learning computing. Software, v2.4.1.
- Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient finetuning of quantized LLMs. *arXiv preprint arXiv:2305.14314*.

- Yuyang Ding, Juntao Li, Pinzheng Wang, Zecheng Tang, Bowen Yan, and Min Zhang. 2024. Rethinking negative instances for generative named entity recognition. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 3461–3475.
- Rong-En Fan, Kai-Wei Chang, Cho-Jui Hsieh, Xiang-Rui Wang, and Chih-Jen Lin. 2008. LIBLINEAR: A library for large linear classification. *Journal of Machine Learning Research*, 9:1871–1874.
- Pierre-Yves Genest, Pierre-Edouard Portier, Előd Egyed-Zsigmond, and Martino Lovisetto. 2025. OWNER: Toward unsupervised open-world named entity recognition. *IEEE Access*, 13:50077–50105.
- Pengcheng He, Jianfeng Gao, and Weizhu Chen. 2023. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. In *The Eleventh International Conference on Learning Representations (ICLR)*.
- Peter Henderson, Jieru Hu, Joshua Romoff, Emma Brunschill, Dan Jurafsky, and Joelle Pineau. 2020. Towards the systematic reporting of the energy and carbon footprints of machine learning. *Journal of Machine Learning Research*, 21:1–44.
- Prathamesh Kalamkar, Astha Agarwal, Aman Tiwari, Smita Gupta, Saurabh Karn, and Vivek Raghavan. 2022. Named entity recognition in Indian court judgments. In *Proceedings of the Natural Language Processing Workshop 2022 (NLLP)*, pages 184–193. Association for Computational Linguistics.
- Vladimir Kaluřev and Branko Brkljač. 2025. Named entity recognition for Serbian legal documents: Design, methodology and dataset development. In *Proceedings of the 15th International Conference on Information Society and Technology (ICIST)*. ArXiv:2502.10582.
- Ioannis Karamitsos, Nikolaos Roufas, Khalil Al-Hussaeni, and Andreas Kanavos. 2025. LegNER: A domain-adapted transformer for legal named entity recognition and text anonymization. *Frontiers in Artificial Intelligence*, 8:1638971.
- Aditya Kusupati, Gantavya Bhatt, Aniket Rege, Matthew Wallingford, Aditya Sinha, Vivek Ramanujan, William Howard-Snyder, Kaifeng Chen, Sham Kakade, Prateek Jain, and Ali Farhadi. 2022. Matryoshka representation learning. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*.
- Alexandre Lacoste, Alexandra Luccioni, Victor Schmidt, and Thomas Dandres. 2019. Quantifying the carbon emissions of machine learning. *arXiv preprint arXiv:1910.09700*.
- Elena Leitner, Georg Rehm, and Julian Moreno-Schneider. 2020. A dataset of German legal documents for named entity recognition. In *Proceedings of the Twelfth Language Resources and Evaluation Conference (LREC)*, pages 4478–4485. European Language Resources Association (ELRA).
- Yi-Te Lu and Yintong Huo. 2025. Financial named entity recognition: How far can LLM go? In *Proceedings of the Joint Workshop of the 9th Financial Technology and Natural Language Processing (FinNLP), the 6th Financial Narrative Processing (FNP), and the 1st Workshop on Large Language Models for Finance and Legal (LLMFinLegal)*, pages 164–168. Association for Computational Linguistics.
- Alexandra Sasha Luccioni, Yacine Jernite, and Emma Strubell. 2024. Power hungry processing: Watts driving the cost of AI deployment? In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*.
- Pedro Henrique Luz de Araujo, Teófilo Emídio de Campos, Renato R. de Oliveira, Matheus Stauffer, Samuel Couto, and Paulo Bermejo. 2018. LeNER-Br: A dataset for named entity recognition in Brazilian legal text. In *Computational Processing of the Portuguese Language (PROPOR)*, volume 11122 of *Lecture Notes in Computer Science*, pages 313–323. Springer.
- Zach Nussbaum, John X. Morris, Brandon Duderstadt, and Andriy Mulyar. 2025. Nomic embed: Training a reproducible long context text embedder. *Transactions on Machine Learning Research (TMLR)*. Published in TMLR 02/2025. OpenReview: <https://openreview.net/forum?id=IPmzyQSiQE>.
- David Patterson, Joseph Gonzalez, Urs Hölzle, Quoc Le, Chen Liang, Lluís-Miquel Munguia, Daniel Rothchild, David So, Maud Texier, and Jeff Dean. 2022. The carbon footprint of machine learning training will plateau, then shrink. *Computer*, 55(7):18–28.
- Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. 2011. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830.
- Qwen Team. 2025. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2020. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21:1–67.
- Vinay N. Ramesh and Rohan Eswara. 2023. FlairNLP at SemEval-2023 task 6b: Extraction of legal named entities from legal texts using contextual string embeddings. In *Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023)*. ArXiv:2306.02182.

Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Oscar Sainz, Iker García-Ferrero, Rodrigo Agerri, Oier Lopez de Lacalle, German Rigau, and Eneko Agirre. 2024. GoLLIE: Annotation guidelines improve zero-shot information-extraction. In *The Twelfth International Conference on Learning Representations (ICLR)*.

Florian Schroff, Dmitry Kalenichenko, and James Philbin. 2015. FaceNet: A unified embedding for face recognition and clustering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

Roy Schwartz, Jesse Dodge, Noah A. Smith, and Oren Etzioni. 2020. Green AI. *Communications of the ACM*, 63(12):54–63.

Emma Strubell, Ananya Ganesh, and Andrew McCallum. 2019. Energy and policy considerations for deep learning in NLP. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*.

Erik F. Tjong Kim Sang and Fien De Meulder. 2003. Introduction to the CoNLL-2003 shared task: Language-independent named entity recognition. In *Proceedings of the Seventh Conference on Natural Language Learning (CoNLL)*.

Nguyen Xuan Vinh, Julien Epps, and James Bailey. 2010. Information theoretic measures for clusterings comparison: Variants, properties, normalization and correction for chance. *Journal of Machine Learning Research*, 11:2837–2854.

Xiang Wei, Xingyu Cui, Ning Cheng, Xiaobin Wang, Xin Zhang, Shen Huang, Pengjun Xie, Jinan Xu, Yufeng Chen, Meishan Zhang, Yong Jiang, and Wenjuan Han. 2023. ChatIE: Zero-shot information extraction via chatting with ChatGPT. *arXiv preprint arXiv:2302.10205*.

Carole-Jean Wu, Ramya Raghavendra, Udit Gupta, Bilge Acun, Newsha Ardalani, Kiwan Maeng, Gloria Chang, Fiona Aga, Jinshi Huang, Charles Bai, et al. 2022. Sustainable AI: Environmental implications, challenges and opportunities. In *Proceedings of Machine Learning and Systems (MLSys)*.

Urchade Zaratiana, Nadi Tomeh, Pierre Holat, and Thierry Charnois. 2024. GLiNER: Generalist model for named entity recognition using bidirectional transformer. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT), Volume 1: Long Papers*, pages 5364–5376. Association for Computational Linguistics.

Wenxuan Zhou, Sheng Zhang, Yu Gu, Muhao Chen, and Hoifung Poon. 2024. UniversalNER: Targeted distillation from large language models for open named entity recognition. In *The Twelfth International Conference on Learning Representations (ICLR)*.

A Full evaluation setup

Table 5: Legal evaluation setup.

Component	Setting
Entity embedder	Nomic v1.5, $D=768$, span-in-context, released checkpoint loaded as-is, not retrained (Nussbaum et al., 2025)
Detector (OPENER-ZS)	GLiNER-L, frozen, threshold 0.3 (Zaratiana et al., 2024)
Detector (OPENER-Sup)	GLiNER-M, frozen, threshold 0.3, probe fit on its own train detections
Zero-shot typing (end-to-end)	English anchor dictionary (1 to 5 words per type), each anchor embedded in 4 short templates and averaged before ℓ_2 normalisation, then 3 transductive spherical k -means iterations and detector fusion $\beta=0.05$
Zero-shot typing (on gold)	plain nearest prototype, single-word anchor embeddings, no transductive pass and no fusion (no detector in the loop)
Supervised typing	LinearSVC one-vs-rest, $C=1.0$, class_weight=balanced, target train labels (Fan et al., 2008)
Baselines	GLiNER-S/M/L (Zaratiana et al., 2024), GNER-T5-base (Ding et al., 2024), OWNER (Genest et al., 2025), Qwen2.5-1.5B-Instruct (Qwen Team, 2025) in NF4 4-bit (Dettmers et al., 2023), all zero-shot on the target type names
Test splits	first 1000 entity-bearing sentences of each test split, Qwen on a 200-sentence subsample
German-LER schema	7-type coarse in the tables, 19-type fine as a stress test
Quality	AMI ($\times 100$), typing-on-gold and end-to-end kept apart
Embedder seeds	7 / 42 / 123, full retraining per seed (42 released)
Test-set bootstrap	$K=1000$, mentions resampled on gold (std ± 0.8 –1.7 AMI), sentences resampled <i>paired</i> end-to-end
Latency / energy	p50 per sentence over warmed-up, GPU-synchronised passes, energy from CodeCarbon normalised per 1000 sentences, released embedder. One consumer GPU (NVIDIA GTX 1660 Ti Max-Q, 6 GB VRAM, CUDA 12.1, PyTorch 2.5.1), same machine for every system, $\approx \pm 15\%$ run-to-run (thermal throttling)
OWNER cost	amortised from wall-clock t : energy = $t \times 110$ W, latency = t /sentences

Only the supervised typing head sees legal labels, every other component is a ready-to-use checkpoint. Detector variants are a free modular choice, so each OPENER head takes its best one. Baselines detect and type jointly, hence the end-to-end comparison only. Bootstrap CIs agree to ± 0.1 AMI across resampling seeds 42 / 7 / 123.